Notes: Allen, Carletti, and Marquez (RFS, 2011)

Credit Market Competition and Capital Regulation

Contents

1 Big picture

- **Question.** Why do banks often hold capital *above* regulatory minimums, and when does credit-market competition make privately chosen capital differ from the socially optimal amount?
- **Setting.** The paper builds a one-period bank-lending model. Firms need outside finance; banks lend and monitor; banks fund themselves with costly capital and deposits; and deposit insurance may or may not be present.
- **Core challenge.** Bank monitoring is unobservable and costly. The bank therefore faces a classic moral-hazard problem. Two instruments can discipline it: a higher loan rate, which raises the bank's upside if the project succeeds, and more bank capital, which makes the bank internalize more downside risk.
- **Main conceptual contribution.** The paper shows that market discipline can come not only from the **liability side** (depositors demanding compensation for bank risk), but also from the **asset side**. In competitive credit markets, borrowers prefer banks that use capital to commit to monitor more.
- **Main baseline result without deposit insurance.** In a perfectly competitive loan market, the market equilibrium features **positive bank capital even without regulation**. Competition pushes banks to choose the highest capital level consistent with full monitoring and zero profits.
- **Main welfare result without deposit insurance.** The market can **overcapitalize** banks relative to the constrained social optimum. The regulator prefers to economize on costly capital and, whenever feasible, provide incentives through a higher loan rate because the loan rate is largely a transfer while capital is a resource cost.
- **Main result with deposit insurance.** Fixed-rate deposit insurance weakens liability-side discipline, but **does not eliminate asset-side discipline**. Competitive banks still voluntarily hold positive capital because borrowers value the monitoring commitment created by capital.
- **Comparison under deposit insurance.** For most of the parameter space, the competitive market still chooses either **too much capital** or the **constrained-efficient** amount. Only in a small region does the market choose too little capital relative to the regulatory benchmark.
- **Key market-structure result.** Monopoly works very differently from competition. Without deposit insurance, a monopolist internalizes the gains from monitoring and the market outcome is constrained efficient. With deposit insurance, monopoly creates the standard moral-hazard problem and a clearer role for capital regulation.
- **Key extension results.**
  - Franchise value and capital are substitute ways to induce monitoring.
  - Capital regulation makes **relationship lending** more attractive relative to transactional lending.
  - Fairly priced deposit insurance reproduces the no-insurance outcome because it restores liability-side discipline.

- **Economic takeaway.** Competition can make banks hold substantial capital voluntarily, but privately chosen capital is not automatically socially optimal. Whether the market wants "too much" or "too little" capital depends on how competition allocates surplus and on whether deposit insurance breaks liability-side pricing of bank risk.

2 Model environment and key objects

2.1 Agents, project, and financing need

A representative firm has a project that requires one unit of investment and pays

$$\begin{cases} R, & \text{if the project succeeds,} \\ 0, & \text{if the project fails.} \end{cases} \tag{1}$$

The firm borrows from a bank and promises a total repayment of r_L if the project succeeds.

Interpretation. The project's scale is normalized to one, so all financing choices can be summarized by the share financed with capital and the share financed with deposits.

2.2 Bank funding and the cost of capital

The bank finances itself with:

- capital k, at per-unit opportunity cost $r_E > 1$, and
- deposits $1 - k$, at normalized opportunity cost 1.

The bank promises depositors a payment r_D.

Interpretation. The key maintained assumption is that equity capital is more expensive than deposits. This is the force that makes the choice between capital and loan-rate incentives economically nontrivial.

2.3 Monitoring technology

The bank chooses unobservable monitoring effort $q \in [0, 1]$, which directly becomes the project's success probability. Monitoring costs are

$$\frac{q^2}{2}. \tag{2}$$

Interpretation. More monitoring is socially valuable because it raises the success probability of the firm's project, but it is privately costly for the bank, so incentives are required.

2.4 Timing

The paper studies two environments.

- **Market case.** The bank first chooses k, then sets r_D and r_L; the firm decides whether to borrow; finally the bank chooses monitoring q.

- **Regulatory case.** A regulator first chooses k; afterward the bank chooses r_D and r_L, the firm decides whether to borrow, and then the bank chooses q.

Interpretation. The regulator controls only capital, not the loan rate. That restriction is central for the welfare comparison.

2.5 Borrower surplus and bank profit

Borrower surplus is

$$BS = q(R - r_L). \tag{3}$$

For given (k, r_D, r_L), bank expected profit is

$$\Pi = q\big(r_L - (1-k)r_D\big) - kr_E - \frac{q^2}{2}. \tag{4}$$

Interpretation. The bank receives the loan repayment only if the project succeeds, repays depositors only in that state, always bears the opportunity cost of capital, and always pays the monitoring cost.

2.6 Monitoring choice

Solving the bank's last-stage problem gives

$$q = \min\{r_L - (1-k)r_D,\ 1\}. \tag{5}$$

Interpretation. Monitoring rises with the loan rate r_L and with capital k, and falls with the deposit rate r_D. Capital and the loan rate are therefore substitute incentive instruments.

2.7 Depositor pricing without deposit insurance

Without deposit insurance, competitive depositors require expected repayment equal to their opportunity cost:

$$qr_D = 1, \qquad \text{so} \qquad r_D = \frac{1}{q}. \tag{6}$$

Interpretation. Better monitoring lowers the promised deposit rate. This is the standard **liability-side discipline** channel.

2.8 Welfare benchmark without deposit insurance

In the no-insurance case, social welfare is just the sum of borrower surplus and bank profit:

$$SW = \Pi + BS = q\big(R - (1-k)r_D\big) - kr_E - \frac{q^2}{2}. \tag{7}$$

Interpretation. The only real resource costs are monitoring and the high opportunity cost of capital. The loan rate largely redistributes surplus between bank and borrower.

2.9 Deposit insurance case

With fixed-rate deposit insurance, depositors are guaranteed repayment and therefore the promised deposit rate becomes

$$r_D = 1. \tag{8}$$

Expected deposit-insurance disbursement is $(1-q)(1-k)$. Hence welfare becomes

$$SW = \Pi + BS - (1-q)(1-k) = qR - (1-k) - kr_E - \frac{q^2}{2}. \tag{9}$$

Interpretation. Deposit insurance removes risk pricing from deposits and shifts part of failure costs onto the insurance fund. That is why private feasibility can survive even when social welfare is negative.

3 Model and equilibrium (all equations + interpretation)

3.1 Competitive market without deposit insurance

In the competitive loan market, the bank chooses (k, r_L, r_D) to maximize borrower surplus:

$$\max_{k, r_L, r_D} BS = q(R - r_L) \tag{10}$$

subject to

$$q = \min\{r_L - (1-k)r_D, 1\}, \tag{11}$$
$$qr_D = 1, \tag{12}$$
$$\Pi = q\big(r_L - (1-k)r_D\big) - kr_E - \frac{q^2}{2} \geq 0, \tag{13}$$
$$BS = q(R - r_L) \geq 0, \tag{14}$$
$$0 \leq k \leq 1. \tag{15}$$

The market equilibrium is:

$$\text{if } R \geq 2 - \frac{1}{2r_E}, \qquad k^{BS} = \frac{1}{2r_E}, \quad r_L = 2 - \frac{1}{2r_E}, \quad r_D = 1, \quad q = 1, \tag{16}$$

with

$$BS = SW = R - \left(2 - \frac{1}{2r_E}\right), \qquad \Pi = 0. \tag{17}$$

If

$$R < 2 - \frac{1}{2r_E}, \tag{18}$$

there is no intermediation.

Interpretation. Borrowers want the lowest loan rate consistent with monitoring and bank participation. Since capital and the loan rate are substitutes, competition drives banks to use as much capital as possible subject to zero profits and full monitoring. This is the paper's central **asset-side discipline** result.

3.2 Regulatory choice without deposit insurance

The regulator chooses capital to maximize welfare but leaves the loan rate to the market:

$$\max_k SW = \Pi + BS, \tag{19}$$

subject to the same feasibility conditions and

$$r_L \in \arg\max_r \; q(R - r). \tag{20}$$

The regulatory solution has three relevant lending regions plus shutdown:

A.1: high project return. If $R \geq 2$, then

$$k^{reg} = 0, \qquad r_L = 2, \qquad r_D = 1, \qquad q = 1, \tag{21}$$

with

$$BS = R - 2, \qquad \Pi = \frac{1}{2}, \qquad SW = R - \frac{3}{2}. \tag{22}$$

Interpretation. When projects are very profitable, a sufficiently high loan rate alone can induce full monitoring. The regulator therefore avoids the costly use of capital.

A.2: intermediate-high project return. If $R_{AB} \leq R < 2$, then

$$k^{reg} = 1 - \frac{R^2}{4} > 0, \qquad r_L = R, \qquad r_D = \frac{2}{R}, \qquad q = \frac{R}{2}, \tag{23}$$

with

$$BS = 0, \qquad \Pi = SW = \frac{R^2}{8} - \left(1 - \frac{R^2}{4}\right) r_E. \tag{24}$$

The cutoff is

$$R_{AB} = \frac{4r_E + 2\sqrt{r_E + 2r_E^2 - 6r_E^3 + 4r_E^4}}{r_E + 2r_E^2}. \tag{25}$$

Interpretation. The regulator pushes the loan rate all the way to the borrower's participation constraint ($r_L = R$), because the loan rate is comparatively cheap from a social perspective. Capital is used only to make that loan-rate-based solution incentive compatible.

B: lower but still profitable project return. If

$$2 - \frac{1}{2r_E} \leq R < R_{AB}, \tag{26}$$

then

$$k^{reg} = \frac{1}{2r_E}, \qquad r_L = 2 - \frac{1}{2r_E}, \qquad r_D = 1, \qquad q = 1, \tag{27}$$

with

$$BS = SW = R - \left(2 - \frac{1}{2r_E}\right), \qquad \Pi = 0. \tag{28}$$

Interpretation. Once project returns are too low to use the loan rate aggressively, the regulator is forced into the same full-monitoring, zero-profit arrangement as the market.

C: low project return. If

$$R < 2 - \frac{1}{2r_E}, \tag{29}$$

there is no intermediation.

3.3 Comparison without deposit insurance

The comparison is sharp:

$$\text{if } R > R_{AB}, \qquad k^{BS} > k^{reg}; \tag{30}$$

$$\text{if } 2 - \frac{1}{2r_E} \leq R < R_{AB}, \qquad k^{BS} = k^{reg}. \tag{31}$$

Interpretation. Competition can induce **inefficiently high capital**. Borrowers like capital because it delivers monitoring without raising the loan rate, but they do not bear the full social cost of that capital. The regulator, in contrast, sees capital as a costly input and would rather use the loan rate whenever the project's payoff is high enough.

3.4 Competitive market with deposit insurance

With fixed-rate deposit insurance, the market still chooses (k, r_L) to maximize borrower surplus, but now $r_D = 1$. The equilibrium always satisfies $\Pi = 0$ and $r_L < R$, so borrowers obtain positive surplus.

Region A: full monitoring. If $R \geq R_{AB}$, then

$$k^{BS} = \frac{1}{2r_E}, \qquad r_L = 2 - \frac{1}{2r_E}, \qquad q = 1, \tag{32}$$

with

$$BS = SW = R - \left(2 - \frac{1}{2r_E}\right). \tag{33}$$

Region B: interior monitoring. If $R < R_{AB}$, then

$$k^{BS} = \left(\frac{\sqrt{2r_E} - \sqrt{2r_E - 3(R-1)}}{3}\right)^2 < \frac{1}{2r_E}, \tag{34}$$

$$r_L = 1 - k^{BS} + \sqrt{2r_E k^{BS}}, \qquad q = \sqrt{2r_E k^{BS}} < 1, \tag{35}$$

with

$$BS = q\big(R - 1 + k^{BS} - q\big), \tag{36}$$

and, using zero profits,

$$SW = qR - q^2 - (1 - k^{BS}). \tag{37}$$

The boundary is piecewise:

$$R_{AB} = \frac{3}{2} - \frac{3}{8r_E} + \frac{r_E}{2} \quad \text{for } r_E < \frac{3}{2}, \tag{38}$$

$$R_{AB} = 3 - \frac{3}{2r_E} \quad \text{for } r_E \geq \frac{3}{2}. \tag{39}$$

Interpretation. Deposit insurance kills the depositors' incentive-pricing role, but competition on the lending side still makes borrowers value highly capitalized banks. The market therefore continues to choose positive capital. The difference relative to the no-insurance case is that now intermediation is always privately feasible, although it can be socially undesirable because the insurance fund pays in failure states.

3.5 Regulatory choice with deposit insurance

When deposit insurance is present, the regulator again chooses capital to maximize social welfare while the market chooses the loan rate. The optimal policy has several regions.

A.

$$k^{reg} = 0, \qquad r_L = \frac{R+1}{2}, \qquad q = 1. \tag{40}$$

B.

$$k^{reg} = 3 - R, \qquad r_L = R - 1, \qquad q = 1. \tag{41}$$

C.

$$k^{reg} = R + 1 - 4(r_E - 1), \qquad r_L = 2(r_E - 1), \qquad q = R - 2(r_E - 1) < 1. \tag{42}$$

D.

$$k^{reg} = 0, \qquad r_L = \frac{R}{2}, \qquad q = \frac{R-1}{2} < 1. \tag{43}$$

E.

$$k^{reg} = \frac{1}{2r_E}, \qquad r_L = 2 - \frac{1}{2r_E}, \qquad q = 1, \qquad \Pi = 0. \tag{44}$$

F. There is no intermediation because social welfare is negative.

Interpretation. The regulator alternates between rate-based incentives, capital-based incentives, and shutdown depending on project profitability R and the cost of capital r_E. The main logic remains the same as before: if the loan rate can do the incentive job cheaply enough, the regulator prefers it; when it cannot, the regulator uses capital; and for sufficiently poor projects the regulator would rather not insure or finance at all.

3.6 Comparison with deposit insurance

The paper summarizes the comparison between the market and regulatory allocations into four regions:

$$\text{A:}\quad k^{BS} > k^{reg}, \tag{45}$$
$$\text{B:}\quad k^{BS} = k^{reg}, \tag{46}$$
$$\text{C:}\quad k^{BS} < k^{reg}, \tag{47}$$
$$\text{D:}\quad \text{no intermediation under regulation.} \tag{48}$$

Interpretation. Even with deposit insurance, the market usually chooses either too much capital or the constrained-efficient amount. Only in a small region does it undercapitalize relative to the regulator. Deposit insurance mainly shifts the boundaries upward because it weakens monitoring incentives and therefore makes capital more valuable as a disciplinary device.

3.7 Extensions

3.7.1 Alternative market structure

With monopoly power, the bank sets $r_L = R$ and captures the surplus itself.

- **No deposit insurance.** The monopolist internalizes both the benefits of monitoring and the funding consequences of risk. The market outcome is constrained efficient, so there is no welfare-improving role for capital regulation.
- **Deposit insurance.** A monopolist would never voluntarily hold capital in the market solution, so capital regulation becomes useful in the familiar moral-hazard way.
- **Intermediate competition.** The more surplus borrowers get, the more capital banks use; the more surplus banks get, the less capital they need.

Interpretation. The paper's overcapitalization result is fundamentally a **competition result**. It depends on borrowers capturing the surplus and preferring capital to loan-rate incentives.

3.7.2 Franchise value

In a repeated version of the model, franchise value FV changes the monitoring condition to

$$q = \min\{r_L - (1-k)r_D + \delta FV, 1\}. \tag{49}$$

With a stationary solution,

$$FV = \frac{\Pi}{1 - q\delta}. \tag{50}$$

Interpretation. Future rents make the bank care more about survival. Franchise value and capital are therefore substitutes in providing monitoring incentives.

3.7.3 Relationship versus transactional lending

For a monitored relationship loan,

$$\Pi_R = q\big(R - (1-k)r_D\big) - kr_E - \frac{q^2}{2}. \tag{51}$$

For a transactional loan with fixed success probability p_T and payoff R_T,

$$\Pi_T = p_T\big(R_T - (1-k)r_D\big) - kr_E. \tag{52}$$

Since

$$\frac{d\Pi_T}{dk} = p_T r_D - r_E < 0, \tag{53}$$

capital reduces the attractiveness of transactional lending.

Interpretation. Minimum capital requirements shift banks toward **relationship lending**. This gives the model a portfolio-composition implication, not just a level-of-capital implication.

3.7.4 Alternative monitoring technology

In a Holmstrom–Tirole style extension, the borrower chooses effort e and monitoring reduces private benefits:

$$e = \min\{(R - r_L) - B(1-q),\, 1\}, \tag{54}$$

while the bank chooses

$$q = \min\{[r_L - (1-k)r_D]B,\, 1\}. \tag{55}$$

Interpretation. The exact formulas change, but the qualitative result remains: capital still helps commit the bank to monitoring, and competition still increases the use of capital when borrowers value monitoring.

3.7.5 Fairly priced deposit insurance

If the bank pays an actuarially fair insurance premium C, then

$$C = \frac{(1-q)(1-k)}{q}. \tag{56}$$

Substituting this into the bank's profit function reproduces the no-insurance expression,

$$\Pi = q\big(r_L - (1-k)r_D\big) - kr_E - \frac{q^2}{2}. \tag{57}$$

Interpretation. Fair pricing restores liability-side discipline. Once the insurance premium reflects default risk, fixed-rate insurance no longer distorts monitoring incentives.

4 What makes the results credible (assumptions, welfare logic, empirical predictions, and limits)

4.1 Why competition creates asset-side discipline

The core wedge is simple:

- borrowers benefit from bank monitoring because it raises success probability;
- borrowers dislike a high loan rate because it directly reduces their payoff;
- borrowers do *not* bear the full private opportunity cost of bank capital.

Hence, when competition forces contracts to maximize borrower surplus, the equilibrium leans heavily on capital to deliver monitoring incentives.

Interpretation. The paper is not saying capital is cheap. It is saying that in competitive lending markets, the party choosing the contract values capital mainly for its commitment effect and does not fully internalize its social cost.

4.2 Why the regulator often wants less capital

The welfare comparison is driven by a sharp distinction:

- a higher loan rate is mostly a **transfer** between borrower and bank;
- higher capital is a **real resource cost** because $r_E > 1$.

Interpretation. When project returns are high enough, the regulator prefers to use the loan rate to induce monitoring and save on costly capital. The market does the opposite because the borrower dislikes the high loan rate.

4.3 What the benchmark does and does not allow the regulator to do

The regulator chooses only k, not r_L.

Interpretation. This is important. If the regulator could also set the loan rate, it would push r_L up to the point where costly capital is minimized. The paper therefore compares the market to a **constrained** regulatory benchmark, not to an unconstrained planner.

4.4 How robust is the mechanism across market structures?

The paper's main mechanism is strongest under competition, but it is not knife-edge.

- Under monopoly without deposit insurance, the bank internalizes monitoring benefits and funding costs, so no extra capital regulation is needed.
- With intermediate market structures, the same logic survives continuously: more borrower surplus means more capital; more bank surplus means less capital.
- Franchise value, relationship lending, and borrower moral hazard all reinforce rather than overturn the basic role of capital as a commitment device.

4.5 Empirical predictions

The model generates clear cross-sectional and comparative-static predictions.

- More competition in the banking sector should be associated with **higher capital ratios**.
- Banks engaged in more monitoring-intensive or relationship lending should be **more capitalized**.
- Firms for which monitoring matters most should prefer to borrow from **better-capitalized banks**.
- Fixed-rate deposit insurance should reduce capital and weaken monitoring.
- Higher capital requirements should shift portfolios away from transactional lending and toward relationship lending.
- Better-capitalized banks should be more likely to expand into foreign markets where information problems are more severe.
- Banks with more concentrated insider ownership or stronger shareholder rights should be more capitalized because the internal agency problem is weaker.

Interpretation. The paper is theoretical, but it is tightly tied to observable objects: competition, capital ratios, lending composition, ownership, and the presence of deposit insurance.

4.6 Main limits

The authors are explicit that the baseline leaves out several forces that matter for regulation.

- The model is **partial equilibrium**: there is no contagion or systemic spillover across banks.
- The baseline abstracts from **asset substitution** and portfolio risk-shifting, although one extension points in that direction.
- The model is **static**, so it cannot explain capital buffers held to avoid future violations of regulatory constraints.
- Banks are identical, and the paper does not study cross-bank heterogeneity in lending technologies or business models.
- Deposit-insurance financing is assumed nondistortionary, which makes public support cheaper than in many realistic environments.

Interpretation. These omitted considerations would often push socially optimal capital *up*, so the paper's result that competitive markets can already overcapitalize banks should be interpreted as a narrow but important benchmark, not a complete theory of bank regulation.

5 Figures and key theoretical results

5.1 Figure 1: timing of the model

Read:

- The figure separates the **market case** from the **regulatory case**.
- In the market case, the bank chooses capital first and then sets deposit and loan rates.
- In the regulatory case, the regulator precommits to capital, but the bank still controls loan pricing.

Economic message. This timing explains why the regulator cannot fully replicate the first best. The regulator can shape incentives through capital, but the loan rate remains market determined.

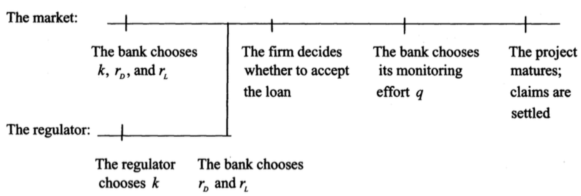

Figure 1
Timing of the model

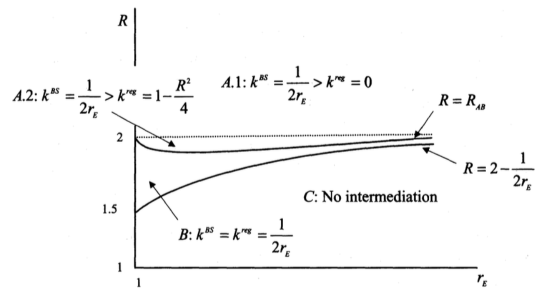

Figure 2
Comparison of market and regulatory solutions with no deposit insurance
The figure compares the level of capital in the market solution (k^{BS}) and in the regulatory solution (k^{reg}) in the case of no deposit insurance as a function of the cost of equity r_E and of the project return R. The figure distinguishes four regions: Region A.1, as defined by $R \geq 2$, where $k^{BS} = 1/2r_E > k^{reg} = 0$; Region A.2, as defined by $R_{AB} \leq R < 2$ with $R_{AB} = (4r_E + 2\sqrt{r_E + 2r_E^2 - 6r_E^3 + fr_E^4})/(r_E + 2r_E^2)$, where $k^{BS} = 1/2r_E > k^{reg} = 1 - R^2/4$; Region B, as defined by $2 - 1/2r_E \leq R < R_{AB}$, where $k^{BS} = k^{reg} = 1/2r_E$; and Region C, as defined by $R < 2 - 1/2r_E$, where there is no intermediation.

5.2 Figure 2: market versus regulation without deposit insurance

Read:

- Region A shows parameter values for which the market chooses $k^{BS} > k^{reg}$.
- Region B shows the constrained-efficient region where $k^{BS} = k^{reg}$.
- Region C shows no intermediation because project returns are too low.

Economic message. This is the cleanest visual statement of the paper's main result: competition can make banks hold **more capital than the regulator wants** even without deposit insurance.

5.3 Figure 3: competitive market with deposit insurance

Read:

- Region A has full monitoring and the same capital formula as in the high-return no-insurance case.
- Region B has interior monitoring, lower capital, and a boundary below which social welfare turns negative.
- The figure highlights that intermediation can remain privately feasible even when $SW < 0$.

Economic message. Deposit insurance weakens liability-side discipline but does not remove the asset-side incentive to capitalize. Borrowers still value capital because it commits the bank to monitor.

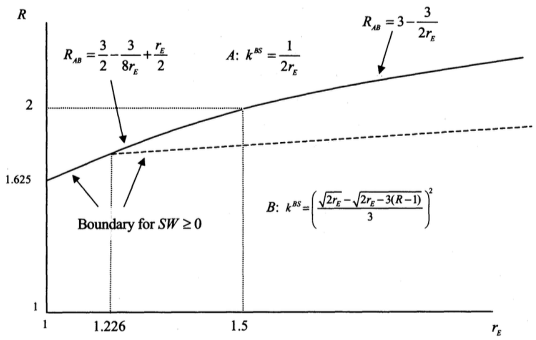

Figure 3
Market solution with deposit insurance
The figure shows the level of capital in the market solution (k^{BS}) for the case of deposit insurance as a function of the cost of equity r_E and of the project return R. The figure distinguishes two regions: Region A, as defined by $R \geq R_{AB}$, where $k^{BS} = 1/2r_E$; and Region B, as defined by $R < R_{AB}$, where $K^{BS} = (\sqrt{2r_E} - \sqrt{2r_E - 3(R-1)}/3)$. The boundary between the two regions is given by $R_{AB} = 3/2 - 3/8r_E + r_E/2$ for $r_E < 3/2$ and by $R_{AB} = 3 - 3/2r_E$ for $r_E \geq 3/2$. The figure also shows the boundary for SW $\geq$ 0. This coincides with R_{AB} for $r_E < 1.266$ and equals R for $r_E \geq 1.266$, where R solves $SW = qR - q^2 - (1 - k^{BS}) = 0$.

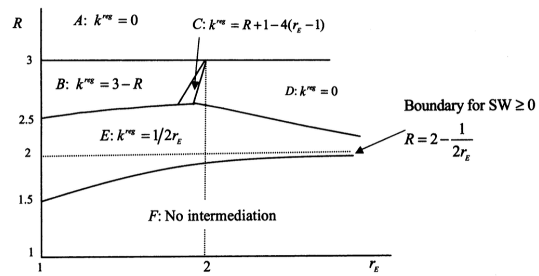

Figure 4
Regulatory solution with deposit insurance
The figure shows the level of capital in the regulatory solution (k^{reg}) for the case of deposit insurance as a function of the cost of equity r_E and of the project return R. The figure distinguishes six regions: Region A, as defined by $R \geq 3$, where $k^{reg} = 0$; Region B, as defined by $R_{BC} \leq R < 3$ and $R_{BE} \leq R < 3$, where $k^{reg} = 3 - R$; Region C, as defined by $R < R_{BC} < 3$, $R_{CE} \leq R < 3$, and $R_{CD} \leq R < 3$, where $k^{reg} = R + 1 - 4(r_E - 1)$; Region D, as defined by $R < R_{CD} < 3$ and $R_{DE} \leq R < 3$, where $k^{reg} = 0$; Region E, as defined by $R_{EF} \leq R$, $R < R_{BE}$, $R < R_{CE}$, and $R < R_{DE}$, where $k^{reg} = 1/2r_E$; Region F, as defined by $R < R_{EF}$, where there is no intermediation. The boundaries between the regions are as follow: $R_{BC} = 2r_E - 1$, $R_{BE} = 3 - 1/2r_E$, $R_{CE} = r_E + \sqrt{r_E - 8r_E^2 + 10r_E^{10} - 3r_E^4}/r_E$, $R_{CD} = 4r_E - 5$, $R_{DE} = (5\sqrt{r_E} + 2\sqrt{3 + r_E})/3\sqrt{r_E}$, and $R_{EF} = 2 - 1/2r_E$. The proof of Proposition 5 contains the intersection points between the boundaries.

5.4 Figure 4: regulatory solution with deposit insurance

Read:

- The figure partitions the parameter space into five lending regions plus a shutdown region.
- In some regions the regulator uses no capital and relies on the loan rate; in others it combines capital and the loan rate; and in Region F it shuts lending down.
- The region boundaries depend jointly on project return R and the cost of capital r_E.

Economic message. Regulation is highly state dependent. Once deposit insurance is present, the regulator may want either more capital, less capital, or no lending at all depending on how costly capital is and how profitable projects are.

5.5 Figure 5: market versus regulation with deposit insurance

Read:

- Region A shows $k^{BS} > k^{reg}$, Region B shows equality, and Region C shows the small area where $k^{BS} < k^{reg}$.

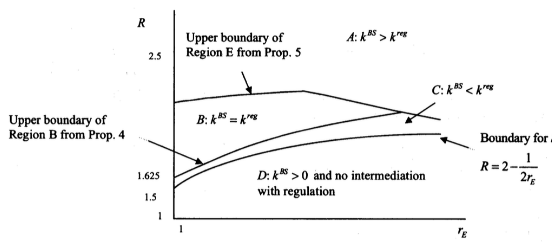

Figure 5
Comparison of market and regulatory solution with deposit insurance
The figure compares the levels of capital in the market solution (k^{BS}) and regulatory solution (k^{reg}) for the case of deposit insurance as a function of the cost of equity r_E and of the project return R. The figure distinguishes four regions: Region A, where $k^{BS} > k^{reg}$; Region B, where $k^{BS} = k^{reg}$; Region C, where $k^{BS} < k^{reg}$; and Region D, where $k^{reg} > 0$ and there is no intermediation with regulation. Region A exists for $R \geq R_{BE} = 3 - 1/2r_E$, $R \geq R_{CE} = r_E + \sqrt{r_E - 8r_E^2 + 10r_E^3 - 3r_E^4}/r_E$, and $R \geq R_{DE} = (5\sqrt{r_E} + 2\sqrt{3 + r_E})/3\sqrt{r_E}$; Region B exists between $R < R_{BE}$, $R < R_{CE}$, $R < R_{DE}$, and $R \geq \hat{R}$, where $\hat{R} = 3/2 - 3/8r_E + r_E/2$ for $r_E < 3/2$ and $\hat{R} = 3 - 3/2r_E$ for $r_E \geq 3/2$. Region C exists for $2 - 1/2r_E \leq R < \hat{R}$; and Region D exists for $R \geq 2 - 1/2r_E$. The proof of Proposition 6 contains the intersection points between the boundaries of Regions A, B, and C.

- Region D shows parameter values where the market would lend but the regulator would rather avoid intermediation.
- Relative to Figure 2, deposit insurance shifts the boundaries upward.

Economic message. The market still usually overcapitalizes or is constrained efficient, but deposit insurance creates a limited undercapitalization region and a larger gap between private and social incentives.

6 Short summary

- Banks hold capital not only because depositors discipline them, but also because borrowers value capital as a commitment to monitor.
- In competitive credit markets, this asset-side discipline leads banks to choose positive capital voluntarily.
- Without deposit insurance, the market often chooses more capital than a constrained welfare-maximizing regulator would choose.
- With deposit insurance, the same basic mechanism survives, though the comparison becomes richer: overcapitalization, constrained efficiency, mild undercapitalization, and socially undesirable lending can all arise.
- The core economic force is that the loan rate is mostly a transfer, while capital is a costly real input.
- Monopoly, franchise value, relationship lending, and fair pricing of deposit insurance all sharpen the interpretation of when capital is and is not needed.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that competition in the loan market can itself induce banks to hold positive capital because capital helps them commit to monitor. This means that observed capital buffers above regulatory minimums need not be explained entirely by regulation, dynamic buffer motives, or liability-side market discipline.

7.2 Economic intuition

Competition hands surplus to borrowers. Borrowers want good monitoring but dislike a high loan rate, so they prefer banks to provide incentives through capital instead. A regulator, however, recognizes that capital is costly in a way the loan rate is not. This difference in objective functions creates the wedge between private and social capital choices.

7.3 Takeaway

The paper's main message is subtle but important: more privately chosen capital is not automatically evidence that the market is doing too little, and it is not automatically evidence that the market is doing the socially right thing either. One needs to understand where market discipline comes from, how competition allocates surplus, and whether deposit insurance breaks the pricing of bank risk.